

Electrochemical Biosensor Immobilization of Formaldehyde Dehydrogenase with Nafion for Determination of Formaldehyde from Indian Mackerel (*Rastrelliger kanagurta*) Fish

NurIndang Marzuki^a, Fatimah Abu Bakar^{a,b,*}, Abu Bakar Salleh^{a,c}, Lee Yook Heng^d, Nor Azah Yusof^e and Shafiquzzaman Siddiquee^f

^aInstitute of Bioscience, Universiti Putra Malaysia, 43400 Serdang, Selangor, Malaysia

^bFaculty of Food Science and Technology, Universiti Putra Malaysia, 43400 Serdang, Selangor, Malaysia

^cFaculty of Biotechnology and Molecular Science, Universiti Putra Malaysia, 43400 Serdang, Selangor, Malaysia

^dFaculty of Science and Technology, Universiti Kebangsaan Malaysia, 43600, Bangi, Selangor, Malaysia

^eFaculty of Science, Universiti Putra Malaysia, 43400 Serdang, Selangor, Malaysia

^fBiotechnology Research Institute, Universiti Malaysia Sabah, JLN UMS, 88400 Kota Kinabalu, Sabah, Malaysia

Abstract: An electrochemical biosensor was developed based on formaldehyde dehydrogenase immobilized with Nafion membrane for determination of formaldehyde in fish. The enzyme was immobilized through the entrapment technique and measured based on the reduction of β -nicotinamide adenine dinucleotide. The response time of the formaldehyde biosensor was <1 min, with an optimum pH of 8. The optimum enzyme loading and NAD^+ concentrations were found at 30 mg/mL and 0.5 mM, respectively. Using the formaldehyde biosensor, a linear response of formaldehyde showed a range of 0.1 to 10 ppm and a detection limit of 0.016 ppm. In application of Nash method, the samples were stored at $4^\circ\text{C} \pm 1$ for 10 days. With the two combined methods, a linear correlation coefficient with $R^2 = 0.9982$ ($y = 0.956x - 0.014$) was found. The developed formaldehyde biosensor showed a good reproducibility, long storage stability (more than 6 months stored at 4°C), and also effective monitoring of formaldehyde level in Indian mackerel (*Rastrelliger kanagurta*) fish.

Keywords: Electrochemical biosensor, Formaldehyde dehydrogenase, Formaldehyde, Nafion, Indian Mackerel.

1. INTRODUCTION

Indian mackerel (*Rastrelliger kanagurta*) are known for the high quality (and are highly harvested) of their meat, which is often very oily. They are an important recreational and commercial fishery. The meat can spoil quickly, especially in the tropics, causing scombroid food poisoning—it should be eaten on the day of capture, unless cured. Fishery in Malaysia is important in providing a source of foods and protein. Mackerel is a very important group of commercial fish tanking second in annual tonnage of marine fishes landed. It is in great demand in fresh condition. About 60% of consumed fresh remain preserved. A large percentage, that is, 40% of the catch is preserved in ice and dispatched to inland areas, and 60% of the catch is salt cured or pickled or sun dried or smoked. They are of importance to many countries bordering the Indian Ocean and the Western Pacific Ocean, for which reason several international integrated research projects are sponsored by organizations such as the Indo Pacific Fisheries Council of the FAO.

Formaldehyde is a flammable, colorless, and readily polymerized gas at ambient temperature, and commercially

available form of 30–50% in aqueous solution [1]. It is a common carbonyl group compound, and is extensively used in consumer goods to protect the products from spoilage by microbial contamination. In the food industry, formaldehyde is often added to keep the food looking fresh, and the consumers are not aware that an excess of this chemical poses a hazard to human health [2,3].

More specifically, formaldehyde is widely used in food processing for its bleaching effect and as a preservative product that prevents food from spoilage by microbial contamination, especially fish and seafood [4]. For this reason, 37% of formaldehyde has been employed as a chemical controlling agent for ectoparasites and aquatic fungi in hatcheries and culture facilities [5,6]. Usually, as described by Cho & Yang [7] and Greg *et al.* [8], formalin has been used to knock down pathogens, likewise preserving fish and shrimp by immersing them in a formaldehyde bath before the boats land on the shore; this happens most likely in a hot climate country where contamination outbreaks easily occur. Nevertheless, its residues in the food for human consumption are prohibited due to possible carcinogenicity of the compound [9].

Human health risks caused by formalin or formaldehyde are well documented, where it may be carcinogenic, anti-genic, and toxic fatal to humans at elevated concentrations [8]. These have been proved by experiments on microorgan-

*Address correspondence to these authors at the Faculty of Food Science and Technology, Universiti Putra Malaysia, 43400 Serdang, Selangor, Malaysia; Tel: 03-8946 8375 / 8376; Fax: 03-8942 3552; E-mails: fatim@putra.upm.edu.my; shafiqpab@ums.edu.my

isms (mutagenic effect) and also on mice and rats (induction of cancer) [10]. An acute ingestion of this chemical by humans has resulted in loss of consciousness, vascular collapse, pneumonia, hemorrhagic nephritis, and abortion. Formaldehyde not only sporadically injures the larynx and trachea but also damages the gastrointestinal tract, which occurs primarily in the stomach and lower oesophagus, as reported by James [11]. It negatively influences the central nervous system, blood, and immune system [12, 13]. This toxic substance may become a reason of growth disorder, blindness, and respiratory diseases of a great number of living organisms [14, 15]. It causes allergic reactions of the respiratory system, eyes, and skin of a human. Formaldehyde is one of the chemical mediators of apoptosis and is included in the category of the carcinogens [14, 16].

Therefore, limits for human exposure and health risk in occupations involving formaldehyde use and standards have been set by the Occupational Safety and Health Administration, as well as by the EPA. In the twentieth century, colorimetric detection methods for formaldehyde such as Deniges' method and Eegriwe's have usually been used [17]. Besides, some standard methods are being used for the quantitative and qualitative analysis of formaldehyde, namely, gas chromatography [17], fluid chromatography [18], fluorometry, refractometry, and spectrophotometry [19–21]. These methods have some serious drawbacks, such as the necessity of using relatively expensive and complicated equipment, the use of toxic reagents, and the presence of highly qualified personnel. Moreover, the major part of them is not sensitive, selective, and specific enough, and not suitable for the express analysis of a great number of samples. Thus, formaldehyde analysis requires the development of some alternative approaches that would be deprived of the majority of drawbacks (ideally—all of them), peculiar to routine methods of its determination. It seems that only bio- and chemo-sensors, a new generation of analytical devices based on electrochemical or optical transducers and selective molecules of biological and chemical origin, merit these requirements.

In recent times, the developments of biosensors have increasingly been investigated, in which the biological recognition molecule is integrated with the physical transducers. Electrochemical-based biosensors enable a direct, reliable, and reproducible measurement, and have general advantages such as real-time measurement, high sensitivity, and selectivity; thus, they have become a good alternative for the determination of formaldehyde [22, 23].

Identification of various types of dehydrogenase where some 250 NADH-dependent dehydrogenases and more than 150 NADPH-dependent enzymes have been found, in turn, initiated the development of dehydrogenase-based sensors [23]. Formaldehyde dehydrogenase is an NAD^+ -dependent enzyme that belongs to the zinc-containing medium-chain alcohol dehydrogenase (ADH) family [24], does not require glutathione, and has been used in the development of formaldehyde biosensors. The specificity of this enzyme toward the substrate, formaldehyde among other aldehyde series [25], has been described. The advantage of using the dehydrogenase enzyme is that molecular oxygen is not only involved in the enzymatic reaction but also does not hinder the result for detection [22].

Immobilization of the enzyme in a membrane through chemical modification could enhance electrocatalytic properties of the electrode as well as the electrochemical performance. The coated electrodes can decrease the fouling by adsorption of surface active agents such as proteins [26,27]. A polymer such as Nafion is cation exchanger has been used to immobilize glucose oxidase for the determination of glucose [28–30]. Nafion is a hydrophobic polyelectrolyte consisting of nonpolar $\text{N} = (\text{CF}_2\text{-CF}_2)$ and polarizable $\text{P} = (\text{O-CF}_2\text{-CF}(\text{CF}_3))_2\text{-CF}_2\text{-CF}_2\text{-SO}_3\text{H}$ segment. [31]. Mauritz and Moore [32] reported that it possesses excellent chemical stability and high affinity toward large organic cations. Thus, Nafion solutions are broadly used in the formation of polymer modified electrodes for an assortment of electroanalytical investigations, respectively.

The aim of the present study is to develop new electrochemical biosensors for the sensitivity detection of formaldehyde using formaldehyde dehydrogenase (FDH) dependent on nicotinamide adenine dinucleotide (NAD^+) as a coenzyme. The enzyme is immobilized in an ion-conducting polymer, which uses Nafion as a membrane by the entrapment technique. The developed formaldehyde biosensor is optimized for several parameters such as pH, ratio of Nafion: FDH, enzyme loading, and concentration of NAD^+ . After optimization, the developed modified electrode is then used to evaluate the formaldehyde level and recovery study in Indian mackerel (*Rastrelliger kanagurta*) fish.

2. MATERIALS AND METHODS

2.1 Reagents

Formaldehyde dehydrogenase [EC 1.2.1.46] from *Pseudomonas putida* (specific activity 1.6 U/mg), β - nicotinamide adenine nucleotide, reduced β - nicotinamide adenine nucleotide, potassium hydrogen phosphate, potassium hydroxide, trichloroacetic acid ACS reagent (99%), and ammonium acetate were purchased from Sigma. Formaldehyde (37%), acetylacetone, and acetic acid were purchased from Merck, and Nafion®117 solutions (5%) were purchased from Fluka. Other chemicals of an analytical grade were received without further purification. All aqueous solutions were prepared with deionized water.

2.2. Indian Mackerel (*Rastrelliger kanagurta*) Sample

Fresh Indian mackerel collected from Sri Serdang wet market was used in this study. The fish samples were stored at $4^\circ\text{C} \pm 1$ for 10 days at an interval of 2 days before being taken for formaldehyde analysis. The fish was thawed, and only the flesh was taken and cut into small portion. For biosensor detection, 15 g of flesh was homogenized together with 30 mL of phosphate buffer (pH 8, 0.1 M) for 5 min, and aliquots were filtered with Whatman 1. The sample was used directly without further extraction and pre-treatment. For the Nash method, the samples were extracted as previously described by Benchmann [33]. A 15 g fish was homogenized with 30 mL of trichloroacetic acid (6%) and filtered as just described. Subsequently, the aliquots were adjusted to pH 6–7 with KOH (15 % w/w) solution and were incubated in ice cubes for 30 min. The recovery tests for formaldehyde were conducted after the addition of a known concentration of formaldehyde (1, 5 and 10 ppm) to the fish sample.

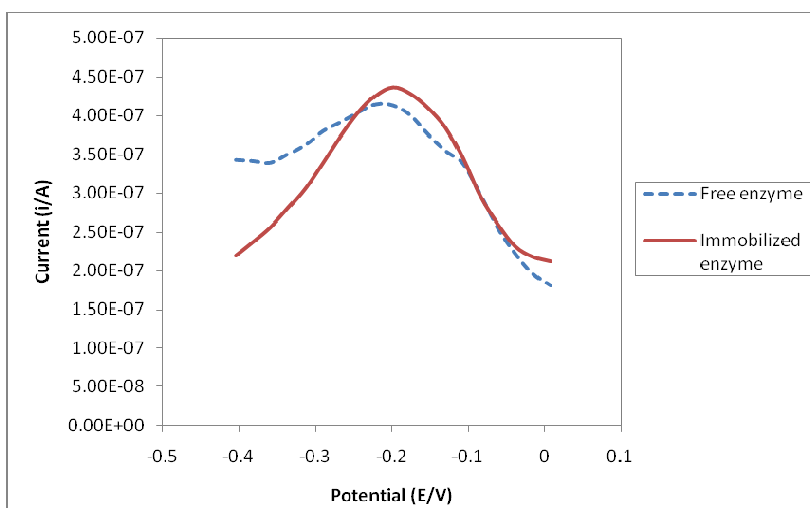


Fig. (1). Differential pulse voltammogram of formaldehyde redox reaction using gold electrode in the presence (—) and absence (---) of Nafion membrane. Potassium phosphate buffer (pH 8, 0.1M), 0.5 mM NAD^+ , 30 mg/ml FDH, $E_{\text{app}} = -0.2$ V.

2.3. Instrumentation

An electrochemical study was carried out using a μ Auto-lab III (Eco-chemie, Netherland) for cyclic voltammetry (CV) and differential pulse voltammetry (DPV) analysis. This device was equipped with electrochemical cells that consisted of a platinum rod as counter electrode (CE), an Ag/AgCl as the reference electrode (RE), and a gold rod as the working electrode (WE). The potentiostat was connected with a DELL personal computer, which allowed the display of output of the electrochemical system using General Purpose Electrochemical System (GPES) software. The solution was purged with high-purity nitrogen to maintain a nitrogen atmosphere. A UV-vis spectrophotometer (Genesys 20, Fisher) was used for the Nash method.

2.4. Preparation of Formaldehyde Biosensor

The gold electrode (AuE) was polished with 0.05 μm alumina slurry for 2 min on a smooth cloth before the electrode was used to ensure that the surface was cleaned from any foreign residues. Then, the electrode was ultrasonicated in an ultrasonic bath for 15 min and dried with a nitrogen stream. Potassium phosphate buffer (0.1 M) was prepared by dissolving 0.34 mg of KH_2PO_4 in 25 mL deionized water, and the pH was adjusted using potassium hydroxide. For immobilization of FDH with Nafion, FDH solution (30 mg/mL) was prepared by dissolving 6 μL of stock FDH solution (50 mg/mL) in 10 μL of 0.1 M phosphate buffer (pH 8), which corresponded to the optimum pH. The FDH solution was thoroughly mixed with the Nafion in the volume ratio of 1: 20 and sonicated for 15 min. Then, 2 μL of the mixture was coated on the surface of a gold electrode and allowed to evaporate at room temperature and subsequently, the thin film covered the gold working electrode. This procedure resulted in the fabrication of homogenously dispersed FDH/Nafion immobilized on the gold electrode surface. For the preparation of NAD^+ solution (0.5 mM), 2.5 μL of NAD^+ solution (5 M) was dissolved in 25 mL of deionized water, whereas 100 ppm formaldehyde stock solution (6.8 μL of 37% formaldehyde in 25 mL deionized water) and standard formaldehyde solution were prepared by adding deionized

water to make stock formaldehyde solution. All the solutions were prepared fresh at the beginning of each experiment.

2.5. Nash Method for Determination of Formaldehyde

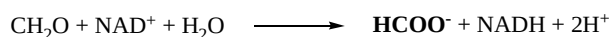
Nash reagent was prepared by dissolving 7.5 g of ammonium acetate, 0.15 mL of acetic acid, and 0.1 mL of acetylacetone, and the volume was adjusted to 50 mL with water. Formaldehyde standards for the Nash test were prepared in a similar manner as for the biosensor determination. For analysis of the formaldehyde level, 2 mL of Nash reagent was added with 1 mL of standard formaldehyde or the sample was put in test tubes and incubated for 30 min at 37°C. Finally, the absorbance reading was measured at a wavelength of 415 nm.

2.6. Statistical Analysis

All the results were performed in triplicate with standard deviation for all the values being <5%. The measurements were calculated using software Minitab version 14. The statistical analysis used to evaluate the result was ANOVA and General Linear Model (GLM).

3. RESULTS AND DISCUSSION

Formaldehyde level is determined by monitoring the reduction of NAD^+ through an enzymatic reaction at a constant potential of -0.2 measured by DPV. An amperometric enzyme-based sensor measurement at 0.2 V vs. Ag/AgCl reference electrode detected the released electrons of the enzyme by means of a redox mediator reported by Achmann et al. [34]. Formaldehyde dehydrogenase acts as the electron transfer that facilitates the addition of one hydrogen atom to NAD^+ and reduces it to NADH, whereas formaldehyde will be converted to formic acid. The enzymatic reaction equations are shown next:



The behavior of the enzyme reaction for free and immobilized enzymes in the presence of formaldehyde can be observed as in Fig. (1). Based on the results, the intensity of

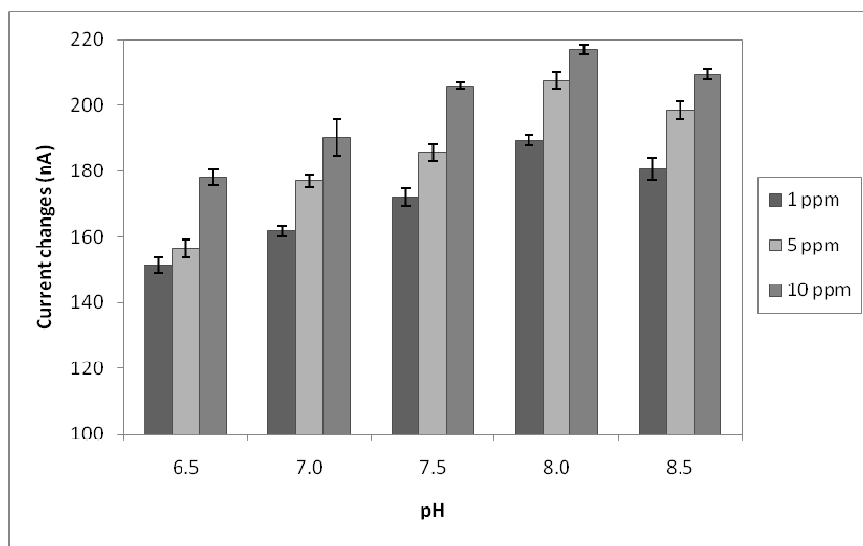


Fig. (2). The pH values for formaldehyde redox reaction using 1, 5, and 10 ppm of formaldehyde in potassium phosphate buffer (pH 8, 0.1 M). 0.5 mM NAD^+ , 30 mg/ml FDH, $E_{\text{app}} = -0.2$ V.

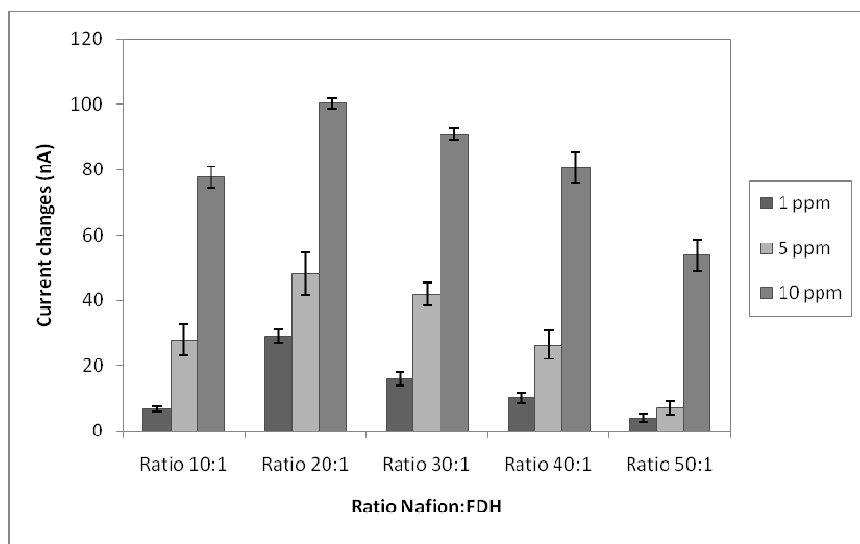


Fig. (3). Effects of Nafion: FDH ratio using 1, 5, and 10 ppm of formaldehyde in potassium phosphate buffer (pH 8, 0.1 M). 0.5 mM NAD^+ . $E_{\text{app}} = -0.2$ V.

cathodic peaks for the immobilized enzyme was higher than those for the free enzyme. This demonstrated that immobilization of the enzyme supported and enhanced the catalytic activity of the enzyme to occur maximally. Busto [35] reported that immobilization of the enzyme minimized denaturation of the enzyme during the operation, improved the stability of the enzyme, and was also found to affect enzyme kinetic changes.

The pH also played an important role in providing the appropriate environment for the enzyme to operate efficiently. However, different pHs may cause the activation or denaturation of the enzyme as well as the enzyme activity. Therefore, this study was conducted by using a free enzyme under its optimum pH range (6.6 to 8.3) to understand the enzyme behavior toward various pH. From the results (refer to Fig. 2), different pH caused large changes to enzyme activity as well as the current changes in the measurement (us-

ing the DPV method). The enzyme activity increased maximally up to pH 8.0 with the increasing of buffer pH but decreased after that. Similar results are also found and reported by Herschkovitz *et al.* [36]. Different pHs have different numbers of electron acceptors and charge propagation [37] through the polymers, which results in the difference in current changes in measurement over the various pH. As the pH increased, the electron diffusion of pH 8.0 also increased and then dropped; after that, the enzyme reaction no longer occurred efficiently due to deprotonation of the ions.

Entrapment of formaldehyde dehydrogenase in Nafion was further explored for its optimum composition of immobilization stability and kinetic properties that was conducted at different ratios. As featured in Fig. (3), the highest current changes in measurement were at ratio 20:1 among five different ratios (10:1, 20:1, 30:1, 40:1, and 50:1) for 1, 5, and 10 ppm of formaldehyde. The study was carried out by vary-

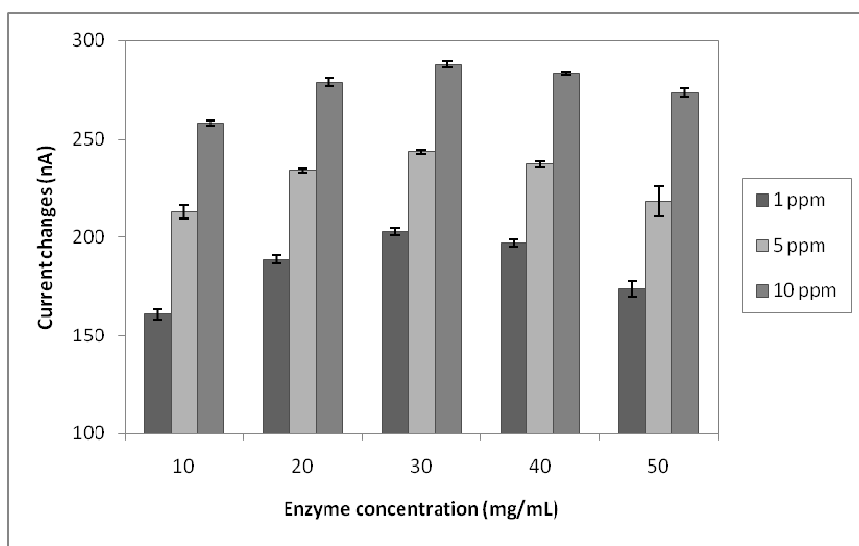


Fig. (4). Effects of different concentrations of formaldehyde dehydrogenase using 0.5 mM NAD^+ in potassium phosphate buffer (pH 8, 0.1 M). $E_{\text{app}} = -0.2$ V.

ing the volume of Nafion, and it maintained the FDH amount, as the enzyme was too expensive to be used in bulk. At an optimum ratio, the FDH was well entrapped homogeneously without interrupting the enzyme structure and polymer properties either chemically or physically and permitting the diffusion of analytes to react maximally. Further, the increase of enzyme activity enhanced the competence of the reaction and decreased the diffusion restriction [38].

Since at a lower ratio, the binding interaction of the Nafion polymer to hold the enzyme was weak, the enzymatic reaction occurred ineffectively. The ratio was increased until it reached 20:1. The declination of the enzyme response at a high concentration of the enzyme, probably due to mass transfer diffusion limitation through the membrane as the protein loading on the membrane, increased. It indicated that the oversaturated protein within the pore space in the Nafion membrane restricted the diffusion of the substrate and the product to react with the enzyme.

Another key parameter which optimized the enzymatic mechanism was varying the enzyme concentration that had been immobilized at an optimum ratio of 20:1 resulting from a previous study. The experiment was carried out at five concentrations of FDH (10, 20, 30, 40, and 50 mg/mL) by using 1, 5, and 10 ppm of formaldehyde as demonstrated in Fig. (4). By using enzyme loading of 30 mg/mL, it exhibited maximal activity of the enzyme by high current in changes measurement but gradually decreased at higher and lower concentrations from the optimum concentration of the enzyme. As explained by Murakami et al. [39], this scenario could be elucidated by the enzyme state during the reaction, where at a higher concentration of the enzyme, the aggregation and autodegradation of the enzyme occurred by itself due to the protein-protein interaction. Elmgren et al. [37] mentioned that at a high concentration, the enzyme within the polymer was limited due to poor assessment of another analyte, as the active site of the enzyme was blocked by other enzymes (compact).

The electron diffusion during the reduction of NAD^+ to NADH should not be hindered among themselves and also within the enzyme itself to work effectively. Further, the reduction of the current response at a high concentration of the enzyme, probably due to mass transfer diffusion limitation through the membrane as the protein loading on the membrane, increased. It also indicated that the oversaturated protein within the pore space in the Nafion membrane restricted the diffusion of the substrate and the product to react with the enzyme. However, at a lower concentration of the enzyme, the capacity of the substrate-enzyme interaction was weak to occur maximally, as the amount of substrate was more than the enzyme molecules entrapped in the Nafion [37].

Fig. (5) shows the effect of NAD^+ concentration on the biosensor response investigated at different concentrations ranging from 0.05 to 2 mM. From the result, 0.5 mM of NAD^+ was found to exhibit an optimum response. This phenomenon could be explained by the excess of NAD^+ molecules in the electrochemical system that have influenced the efficiency of the enzymatic reaction and possibly blocked the active site of the enzyme.

The developed biosensor was further investigated for response range characterization by employing different concentrations of formaldehyde ranging from 0.1 to 20 ppm of formaldehyde (Fig. 6). The linear regression equation for formaldehyde was calculated as $y = 7.0264x + 401.2$, ($R^2 = 0.9865$). Limit of detection (LOD) can be defined as signal-to-noise (S/N) ratios, where it was calculated as the ratio of the response (with formaldehyde) divided by the noise level of the system (without formaldehyde). It could also be determined by using the formula $3N/\text{sensitivity}$ described by Ballantine et al. [40]. The value was found to be 0.016 ppm of formaldehyde. However, the sensitivity of the formaldehyde biosensor can be attained from the slope, which is $7.0264 \text{ nA ppm}^{-1}$.

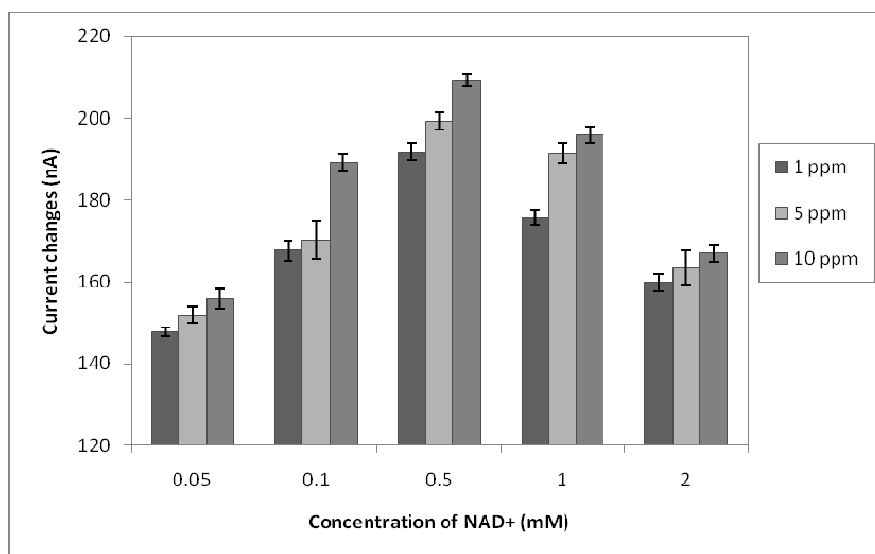


Fig. (5). Effects of different concentrations of NAD⁺. Enzyme loading 30 mg/ml in potassium phosphate buffer (pH 8, 0.1 M) for 1, 5, and 10 ppm formaldehyde. $E_{app} = -0.2$ V.

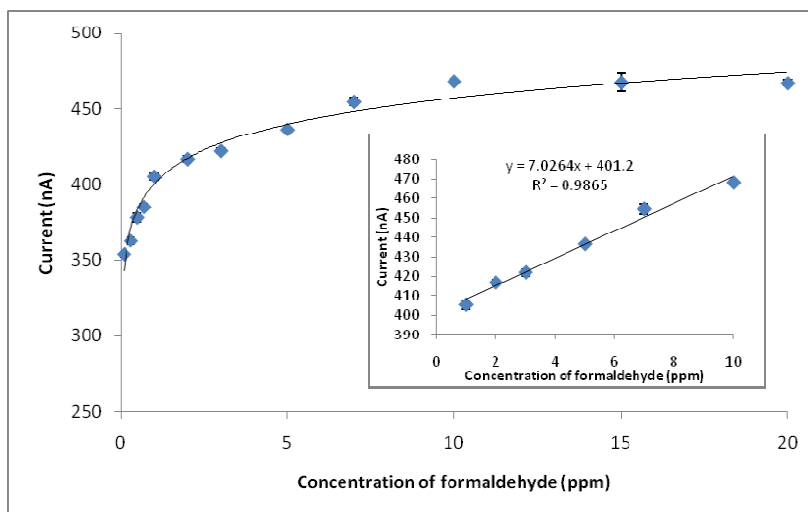


Fig. (6). Response range of formaldehyde where FDH immobilized in Nafion, which produced a linear response. 0.5 mM NAD⁺, enzyme loading 30 mg/mL in potassium phosphate buffer (pH 8, 0.1 M). $E_{app} = -0.2$ V.

The developed electrochemical biosensor showed a good reproducibility, where from the statistical analysis, the results exhibited no significant differences at the 0.05 probability level with a p -value for 1 ppm (0.988), 5 ppm (0.995), and 10 ppm (1.000) ($n=10$). The biosensor responses were found to be stable by retaining 90% of their initial activity even after six months of storage at 4°C. There was a slight decrease in the biosensor response, possibly due to the changes in the enzyme immobilization composition and the loss of the protein chain of enzyme flexibility.

For the interferences study, the formaldehyde biosensor retained its specificity toward formaldehyde, and the addition of methanol or ethanol in the electrochemical system did not give a full-size impact to the response. This was illustrated by the percentage of formaldehyde recovered, which gave 99.3 to 99.8 % (methanol) and 99.0 to 99.8 % (ethanol). This phenomenon can be explained by the properties of the enzyme itself, where the catalytic site of FDH was very spe-

cific and selective to formaldehyde and only caught that substrate in reacting.

For validation of developed formaldehyde biosensors, the results were compared with the conventional Nash method. Comparable results with a good agreement between both methods (Fig. 7) at a 95 % confidence level were obtained with a linear correlation coefficient of ($y = 0.956x - 0.014$, $R^2 = 0.9982$). On the initial day, it was found that the fish contained 0.09 ppm of formaldehyde, which is higher than the results from the Nash method (0.05 ppm), indicating the formaldehyde biosensor is more sensitive, as it used the specific enzyme to interact with the substrate rather than the conventional Nash method, which depended on a chemical reaction. The highest concentration of formaldehyde was observed at day two; then, it gradually declined due to inhibition of TMAOase by the existing formaldehyde as its product to generate the substance.

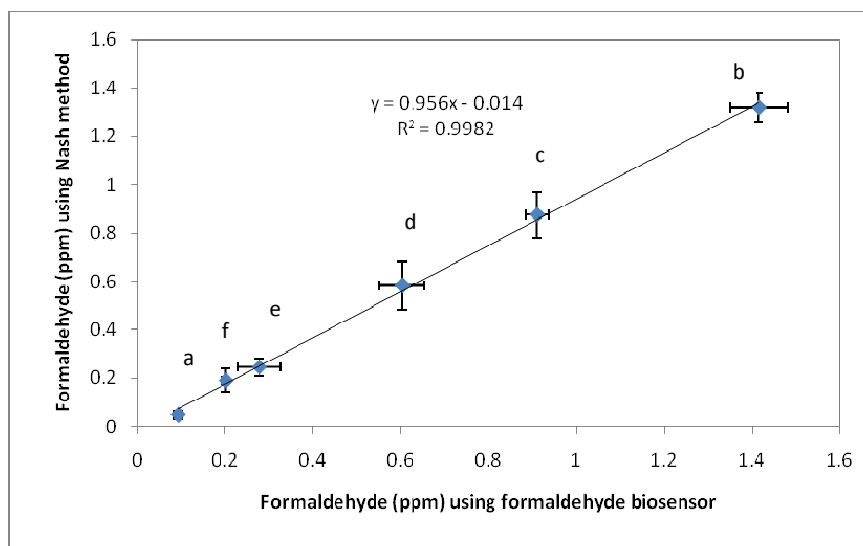


Fig. (7). Correlation of formaldehyde levels determined in fish samples with both formaldehyde biosensors and Nash method (Storage at $-4 \pm 1^\circ\text{C}$: (a = 0 day, b = 2 day, c = 4 day, d = 6 day, e = 8 day, f = 10 day).

Table 1. The Recovery Performance of a Formaldehyde Biosensor for the Analysis of Formaldehyde in Indian Mackerel Samples (Wet Weight)

Spike (ppm)	Duration (day)	Expected Value (ppm)	Found Value (ppm)	FA Recovered (%)	RSD (%)
1	0	1.05	0.89 ± 0.05	85.04	5.31
	2	2.32	1.76 ± 0.04	75.72	2.31
	4	1.88	1.28 ± 0.08	68.22	6.34
	6	1.58	0.94 ± 0.15	59.59	15.95
	8	1.25	0.59 ± 0.08	47.53	13.84
	10	1.19	0.48 ± 0.05	35.83	11.54
5	0	5.05	3.60 ± 0.39	71.32	10.81
	2	6.32	4.18 ± 0.04	66.09	0.86
	4	5.88	2.69 ± 0.06	45.59	2.22
	6	5.58	2.05 ± 0.11	38.65	4.94
	8	5.25	1.91 ± 0.18	38.75	8.81
	10	5.19	1.85 ± 0.04	35.51	1.96
10	0	10.05	6.86 ± 0.35	68.30	5.17
	2	11.32	8.08 ± 0.01	71.36	0.17
	4	10.88	5.39 ± 0.05	49.55	0.91
	6	10.58	4.40 ± 0.08	41.54	1.73
	8	10.25	3.51 ± 0.11	34.30	3.08
	10	10.19	3.41 ± 0.07	33.48	2.08

Note: mean $\pm$ standard deviation (n=3)

The recovery study of standard formaldehyde solution was carried out to observe the effect of the addition of formaldehyde in the electrochemical system; it will either produce the same response as expected or not. The study was performed using coated electrode and was analyzed by the DPV method. After that, the current changes were converted into the concentration of formaldehyde (ppm) unit for a clear understanding. Besides, the percentage of formaldehyde recovered for three concentrations of formaldehyde (1, 5, and

10 ppm) were calculated and conducted in ten days as shown in Table 1. From the results, it was found that after addition of formaldehyde the response was less than the expected value. This may be due to the properties of formaldehyde that is easily volatile, or the enzymatic reaction was slowed down after reacting with a previous substrate and the product could also contribute to the disturbance of the response and, thus, influence the sensor results. However, it showed high formaldehyde recovering between 80 and 98 percent and,

therefore, it is still reliable for the determination of formaldehyde. However, the percentage of formaldehyde recovered was decreased as the storage time increased for both methods due to denaturation of the fish muscles, which affected the reactivity of formaldehyde [41,42].

4. CONCLUSION

Formaldehyde dehydrogenase is successfully immobilized in Nafion membrane onto gold electrode surfaces. The developed formaldehyde biosensor can detect as low as 0.016 ppm of formaldehyde in aqueous solution and a response time of less than 1 min. The formaldehyde-selective sensor shows a good reproducibility, high sensitivity, selectivity, and long storage stability toward formaldehyde. For evaluation in real samples, formaldehyde biosensors showed a good technique for determination of the formaldehyde level in Indian mackerel comparable to the conventional Nash method. The attractive electrochemical and structural properties of Nafion combining with formaldehyde dehydrogenase suggest potential application of this biosensor for the determination of formaldehyde, especially in the food industry.

It also exhibited reproducibility, repeatability, and operational stability dependent on the concentration of formaldehyde.

CONFLICT OF INTEREST

Declared none.

ACKNOWLEDGMENTS

The authors are grateful for the financial support provided by the Malaysian government through the Ministry of Science, Technology, and Innovation (MOSTI) and Malaysian Genome Institute (MGI) under research project no. 001-002-0027.

REFERENCES

- [1] WHO, Formaldehyde, Environmental Health Criteria. Geneva, Switzerland (1989).
- [2] Cui, X.; Fang, G.; Jiang, L.; Wang, S. Kinetic spectrophotometric method for rapid determination of trace formaldehyde in foods. *Anal. Chim. Acta*, **2007**, *590*, 253-259.
- [3] NurIndang, M.; Abdulmir, A.S.; Abu Bakar, F.; Salleh, A.B.; Lee, Y.H.; NorAzah, Y. A review: methods of determination of health-endangering formaldehyde in diet. *Res. J. Pharmacol.* **2009**, *3*, 31-47.
- [4] Wang, S.; Cui, X.; Fang, G. Rapid determination of formaldehyde and sulphur dioxide in food products and Chinese herbals. *J. Food Chem.* **2007**, *103*, 1487-1493.
- [5] Rach, J.J.; Howe, G.E.; Schreier, T.M. Safety of formalin treatments on warm-and cool water fish eggs. *Aquaculture* **1997**, *149*, 183-191.
- [6] Schnick, R.A. The impetus to register new therapeutants for aquaculture. *Prog. Fish-Culture* **1988**, *50*, 190-196.
- [7] Cho, J.K.; Yang, H.C. Determination of formaldehyde residue and histological observation in formalin and neutral-formalin treated Korean rockfish (*Sebasteschlegeli*). *J. Fish Pathol.* **1996**, *9*, 157-168.
- [8] Greg, A.W.; Mertinez, L.M.; Ohara, D.S.; Brower, P.R. Human health risks association with formaldehyde treatment used in aquaculture: Pilot study, 28th annual eastern fish health workshop. 2003.
- [9] Jung, S.H.; Kim, J.W.; Jeon, I.G.; Lee, Y.H. Formaldehyde residues in formalin-treated olive flounder (*Paralichthysolivaceus*), black rockfish (*Sebasteschlegeli*), and seawater. *Aquaculture*. **1991**, *194*, 253-262.
- [10] Squire, R.A.; Cameron, L.L. An analysis of potential carcinogenic risk from formaldehyde. *Reg. Toxicol. Pharmacol.* **1984**, *4*, 107-129.
- [11] James, E.G. Formaldehyde toxicity, Mac Graw Hill, London, 1983.
- [12] Coggon, D.; Harris, E.C.; Poole, J.; Palmer, K.T. Extended follow-up of a cohort of British chemical workers exposed to formaldehyde. *J. Natl. Cancer Inst.* **2003**, *95*, 1608-1615.
- [13] Hauptmann, M.; Lubin, J.H.; Stewart, P.A.; Hayes, R.B.; Blair, A. Mortality from lymphohematopoietic malignancies among workers in formaldehyde industries. *J. Natl. Cancer Inst.* **2003**, *95*, 1615-1623.
- [14] Katakya, R.; Bryce, M.R.; Goldenberg, L.; Hayes, S.; Nowak, A. A biosensor for monitoring formaldehyde using a new lipophilic tetrathiafulvalene-tetracyanoquinodimethane salt and a polyurethane membrane. *Talanta* **2002**, *56*, 451-458.
- [15] International Agency for Research on Cancer (IARC) (1995) Formaldehyde. In: IARC monographs on the evaluation of carcinogenic risks to humans, vol 62, Wood Dust and Formaldehyde, Lyon, pp 217-362.
- [16] Sexton, K.; Petreas, M.X.; Liu, K. Formaldehyde exposure inside mobile homes. *Environ. Sci. Technol.* **1989**, *23*, 985-988.
- [17] Dumas, T. Determination of formaldehyde in air by gas chromatography. *J. Chromatogr.* **1982**, *247*, 289-295.
- [18] Mann, B.; Grajeski, M.L. New chemiluminescentderivatizing agent for the analysis of aldehydes and ketones by high performance liquid chromatography with peroxoxalate chemiluminescence. *J. Chromatogr.* **1987**, *386*, 149-158.
- [19] West, P.W.; Sen, B. Spectrophotometric determination of traces of formaldehyde. *Fresenius' J. Anal. Chem.* 1956, *153*, 177-183.
- [20] Mohimann, G.R. Formaldehyde detection in air by laser induced fluorescence. *Appl. Spectrosc.* **1985**, *39*, 98-101.
- [21] Vianello, F.; Stefani, A.; Di Paolo, M.L.; Rigo, A.; Lui, A.; Margesin, B.; Zen, M.; Scarpa, M.; Soncini, G. Potentiometric detection of formaldehyde in air by an aldehyde dehydrogenase FET. *Sens. Actuators B: Chem.* **1996**, *37*, 49-54.
- [22] Nikitina, O.; Shleev, S.; Gayda, G.; Demkiv, O.; Gonchar, M.; Gorton, L.; Csöregi, E.; Nistor, M. Bi-enzyme biosensor based on NAD⁺- and glutathione-dependent recombinant formaldehyde dehydrogenase and diaphorase for formaldehyde assay. *Sens. Actuators B: Chem.* **2007**, *125*, 1-9.
- [23] Lobo, M.J.; Miranda, A.J.; Tuñón, P. Amperometric biosensors based on NAD(P)-dependent dehydrogenase enzymes. *Electroanalysis* **1997**, *9*, 191-202.
- [24] Tanaka, N.; Kusakabe, Y.; Ito, K.; Yoshimoto, T.; Nakamura, K.T. Crystal structure of formaldehyde dehydrogenase from *Pseudomonas putida*: the structural origin of the tightly bound cofactor in nicotinoprotein dehydrogenases. *J. Mol. Biol.* **2002**, *324*, 519-533.
- [25] Ogushi, S.; Ando, M.; Tsuru, D. Substrate specificity of formaldehyde dehydrogenase from *Pseudomonaseputida*. *J. Agric. Biol. Chem.* **1984**, *48*, 597-601.
- [26] Sittampalam, G.; Wilson, G.S. Surface-modified electrochemical detector for liquid chromatography. *Anal. Chem.* **2002**, *55*, 1608-1610.
- [27] Wang, J.; Hutchins, L.D. Thin-layer electrochemical detector with a glassy carbon electrode coated with a base-hydrolyzed cellulosic film. *Anal. Chem.* **2002**, *57*, 1536-1541.
- [28] Mizutani, F.; Yabuki, S.; Katsura, T. Amperometric enzyme electrode with fast response to glucose using a layer of lipid-modified glucose oxidase and Nafion anionic polymer. *Anal. Chim. Acta* **1993**, *274*, 201-207.
- [29] Rahman, M.M.; Umar, A.; Sawada, K. Development of amperometric glucose biosensor based on glucose oxidase co-immobilized with multi-walled carbon nanotubes at low potential. *Sens. Actuators B: Chem.* **2009**, *137*, 327-333.
- [30] Xu, J.-J.; Yu, Z.-H.; Chen, H.-Y. Glucose biosensors prepared by electropolymerization of p-chlorophenylamine with and without Nafion. *Anal. Chim. Acta* **2002**, *463*, 239-247.
- [31] Jang, S.S.; Molinero, V.; Çagin, T.; Goddard, W.A. Effect of monomeric sequence on nanostructure and water dynamics in Nafion 117. *Solid State Ionics* **2004**, *175*, 805-808.
- [32] Mauritz, K.A.; Moore, R.B. State of understanding of Nafion. *Chem. Rev.* **2004**, *104*, 4535-4586.

- [33] Benchmann, I.E. Determination of formaldehyde in frozen fish with formaldehyde dehydrogenase using flow injection system with an incorporated gel-filtration chromatography column. *Anal. Chim. Acta* **1996**, 320, 155-164.
- [34] Achmanna, S.; Hermann, M.; Hilbrig, F.; Jérôme, V.; ammerle, M.H.; Freitag, R.; Moos R. Direct detection of formaldehyde in air by a novel NAD⁺- and glutathione-independent formaldehyde dehydrogenase-based biosensor. *Talanta* **2008**, 75, 786-791.
- [35] Busto, M.D. An experiment illustrating the effect of immobilisation on enzyme properties. *Biochem. Edu.* **1998**, 26, 304-308.
- [36] Herschkovitz, Y.; Eshkenazi, I.; Campbell, C.E.; Rishpon, J. An electrochemical biosensor for formaldehyde. *J. Electroanal. Chem.* **2000**, 491, 182-187.
- [37] Elmgren, M.; Lindquist, S-E.; Sharp, M. Charge propagation through a redox polymer film containing enzymes-effects of enzyme loading, pH and supporting electrolyte. *J. Electroanal. Chem.* **1993**, 362, 227-235.
- [38] Tarhan, L.; Telefoncu, A. Effect of enzyme ratio on the properties of glucose oxidase and catalase immobilized on modified perlite. *Proc. Biochem.* **1992**, 21, 1-15.
- [39] Murakami, Y.; Hoshi, R.; Hirata, A. Characterization of polymer-enzyme complex as a novel biocatalyst for nonaqueous enzymology. *J. Mole. Catalysis B: Enzym.* **2003**, 22, 79-88.
- [40] Ballantine, D.S.; White, R.M.; Martin, S.J.; Ricco, A.J.; Zellers, E.T.; Frye, G.C.; Wohltjen, H. Chemical and biological sensor. In *acoustic wave sensors: theory, design, and physico-chemical applications*, Levy, M. and Stern, R. eds. San Diego, California: Academic Press, **1997**, pp. 222-320.
- [41] Sotelo, C.G.; Aubourg, S.P.; Perez-Martin, R.I.; Gallardo, J.M. Protein denaturation in frozen stored hake (*Merlucciusmerluccius* L.) muscle: the role of formaldehyde. *Food Chem.* **1994**, 50, 267-275.
- [42] Sotelo, C.G.; Piñeiro, C.; Pérez-Martín, R.I. Denaturation of fish proteins during frozen storage: role of formaldehyde. *Z. Lebensm. Unters. Forsch.* **1995**, 200, 14-23.